Quantum Squeezing and Feature Dimensionality in a Kerr Photonic Frequency-Multiplexed Extreme Learning Machine

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(Dated: May 11, 2026)

We investigate how the quadrature statistics of the optical input state shape the learning capacity of a frequency-multiplexed extreme learning machine (ELM) whose nonlinear mixing layer is provided by the Kerr four-wave-mixing (FWM) interaction. Five input geometries are compared across squeezing parameters $r \in [0, 2]$: coherent, amplitude-squeezed, phase-squeezed, two-mode EPR, and thermal states. A N=32-mode optical frequency comb with nonlinearity strength $\gamma L = 0.5$ is used to evaluate each geometry on three benchmark tasks that probe linearly separable (Iris), nonlinearly separable (XOR Spiral), and temporal-memory (NARMA-10) structure. Phase-squeezed inputs are shown to systematically increase the effective rank of the feature Gram matrix and the centered kernel alignment (CKA) with the task structure, producing a 27.5% improvement in 1−NRMSE over the coherent baseline on NARMA-10 at r=2. Two-mode EPR squeezing suppresses feature diversity through inter-mode correlations. These results establish a direct mapping between the Wigner-function geometry of the input light and the representational capacity of Kerr-photonic learning machines, and provide a practical guide for the design of quantum-enhanced optical neural networks.

INTRODUCTION

Physical dynamical systems are attractive candidates for computational substrates [1, 2]. A physical reservoir is a driven nonlinear system whose transient response is read out by a linear decoder, performing inference at the speed and energy budget of the underlying physical process. Among candidate substrates, photonic platforms are particularly compelling: optical fields propagate without resistive loss, interference and nonlinear mixing implement the high-dimensional feature maps that underpin modern machine learning, and integrated photonics fabrication has matured to the point where wafer-scale neural network chips are commercially available [3, 4].

Kerr Nonlinearity

The Kerr effect provides a natural photonic nonlinearity for photonic computing. In a Kerr medium such as an optical fiber, a microresonator, or a waveguide, the instantaneous refractive index $n(I) = n_0 + n_2 I$ depends on intensity, coupling the phase of each frequency component to the power of all others through self-phase modulation (SPM) and cross-phase modulation (XPM) [5]. This intensity-dependent phase evolution is the origin of all nonlinear mixing in the frequency-comb setting studied here and is described quantitatively in future sections.

Four-Wave Mixing

For a multi-frequency input, the Kerr intensity dependence generates four-wave mixing (FWM): energy is redistributed among comb lines via processes of the form $\omega_l + \omega_n \rightarrow \omega_m + \omega_k$ wherever the energy-conservation condition

$$\omega_l - \omega_m + \omega_n = \omega_k \tag{1}$$

is satisfied [5]. The process is illustrated schematically in Fig. 1: two pump photons at ω_l and ω_n are absorbed and two signal photons at ω_m and ω_k are emitted, with the output intensities $|E_m^{\text{out}}|^2$ and $|E_k^{\text{out}}|^2$ growing quadratically with propagation distance in the low-conversion limit. Because $|E_k|^2$ is quadratic in the input field amplitudes, the full set of FWM output intensities constitutes a degree-four polynomial feature map in the inputs [6].

Extreme Learning Machines

Extreme learning machines (ELMs) are a minimal and analytically tractable realization of the reservoir-computing idea [7]. A fixed, randomly initialized nonlinear projection maps input data into a high-dimensional feature space, and only a final linear readout layer is trained. The computational power of the ELM rests on the phase-space dimensionality of this projection: when the projected features span a sufficient number of dimensions relative to the complexity of the target function, the linear readout can solve tasks that are impossible for any purely linear model. Photonic ELMs of this kind have been demonstrated experimentally in optical fiber [8, 9], semiconductor microresonators [10, 11], and integrated

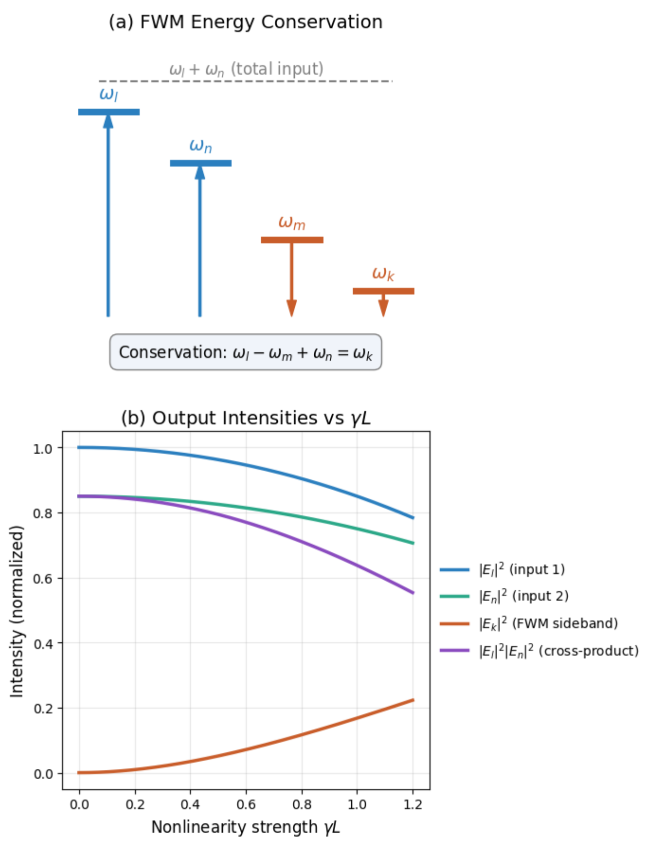

FIG. 1. (a) FWM energy conservation: two input photons (blue, ω_l, ω_n) are absorbed and two output photons (orange, ω_m, ω_k) are emitted subject to the conservation condition $\omega_l - \omega_m + \omega_n = \omega_k$ [5]. (b) Normalized output intensities of the input modes, the FWM-generated sideband $|E_k|^2$, and the cross-intensity product $|E_l|^2|E_n|^2$ as a function of nonlinearity strength γL in the perturbative Kerr regime.

Si_3N_4 chips [3]. Theoretical analyses have identified the mode count N, the nonlinear phase accumulation γL, and the coherence of the input as key design parameters [6, 12].

Quantum noise as a design parameter

In prior studies, the quantum noise of the input field is treated as a fixed, passive background at the minimum shot-noise floor of a coherent laser state. Modern quantum-optical laboratories routinely produce squeezed states, in which the uncertainty is redistributed between the two field quadratures $\hat{X}_1 = (\hat{a} + \hat{a}^\dagger)/2$ and $\hat{X}_2 = (\hat{a} - \hat{a}^\dagger)/(2i)$, with one quadrature below the shot-noise limit at the expense of the other [13, 14]. Squeezed light is an established resource in quantum sensing [15], quantum communication [16], and measurement-based quantum computing, but its role as an active design parameter for photonic machine learning has not been studied. Because the FWM nonlinearity mixes both field quadratures, any asymmetry in their variances directly alters the statistical distribution of the polynomial output features and, consequently, the rank structure of the feature kernel. The central question of this paper is whether the orientation of the quantum noise ellipse affects the learning performance of a Kerr-FWM ELM, and if so, by how much and through what mechanism.

This Letter presents a systematic numerical study of a Kerr-FWM ELM driven by five qualitatively distinct Gaussian input states. The squeezing parameter r is swept from 0 to 2 (up to $\approx$17 dB of noise reduction), and the ELM is evaluated on three benchmark tasks of increasing structural difficulty. The central result is that phase-squeezed inputs provide the largest and most robust gains, particularly on time-series tasks with long memory depth, while two-mode EPR squeezing degrades performance by introducing classical inter-mode correlations.

MODEL

As illustrated in Fig. 2, the physical pipeline of the Kerr-FWM ELM consists of four stages: data encoding via coherent displacement, quantum noise injection, nonlinear mixing in a Kerr medium, and feature extraction via optical measurement. The physics of each subsystem is detailed below.

Gaussian-state optical frequency comb

The input to the ELM is an N-mode optical frequency comb in which each comb line $k = 1, \ldots, N$ carries an independent, Gaussian-distributed complex field amplitude. This models the output of an optical parametric oscillator or a squeezed-light source injected into a microresonator [13, 16]. The amplitude of mode k is written as

$$E_k = \alpha_k + \xi_k^{(1)} + i\,\xi_k^{(2)}, \quad (2)$$

where $\alpha_k \in \mathbb{R}$ is the coherent displacement encoding the input data, and $\xi_k^{(1,2)}$ are zero-mean Gaussian quadrature noise variables. For a Gaussian state the Wigner function is a two-dimensional Gaussian in the (X_1, X_2) phase space [13]. The vacuum shot-noise level is set to $\sigma^2_{\text{vac}} = 1/2$ per quadrature, consistent with the canonical commutation relation $[\hat{X}_1, \hat{X}_2] = i/2$.

The Wigner phase-space distributions of the five input geometries are shown in Fig. 3, and their quadrature variances are collected in Table I.

Coherent state. Both quadrature variances equal the vacuum level, $\text{Var}(\xi^{(1)}) = \text{Var}(\xi^{(2)}) = 1/2$. The Wigner

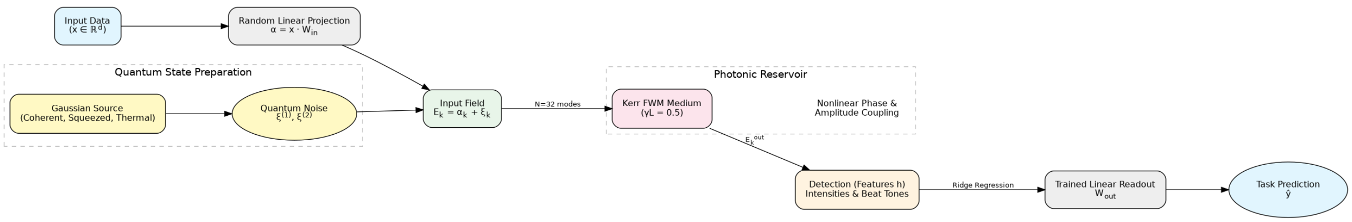

FIG. 2. Physical pipeline of the Kerr-FWM extreme learning machine. (1) Classical input data $\mathbf{x}$ are encoded into the coherent displacement α_k of a frequency comb. (2) Gaussian quantum noise of a specified squeezing geometry $\boldsymbol{\xi}$ is injected, defining the initial multimode state. (3) The optical field propagates through a Kerr nonlinear medium (optical fiber or microresonator), acquiring nonlinear phase and FWM amplitude transfer. (4) Photodetection and homodyne measurement extract first- and second-order intensity moments as classical features $\mathbf{h}$, which train the ridge-regression readout.

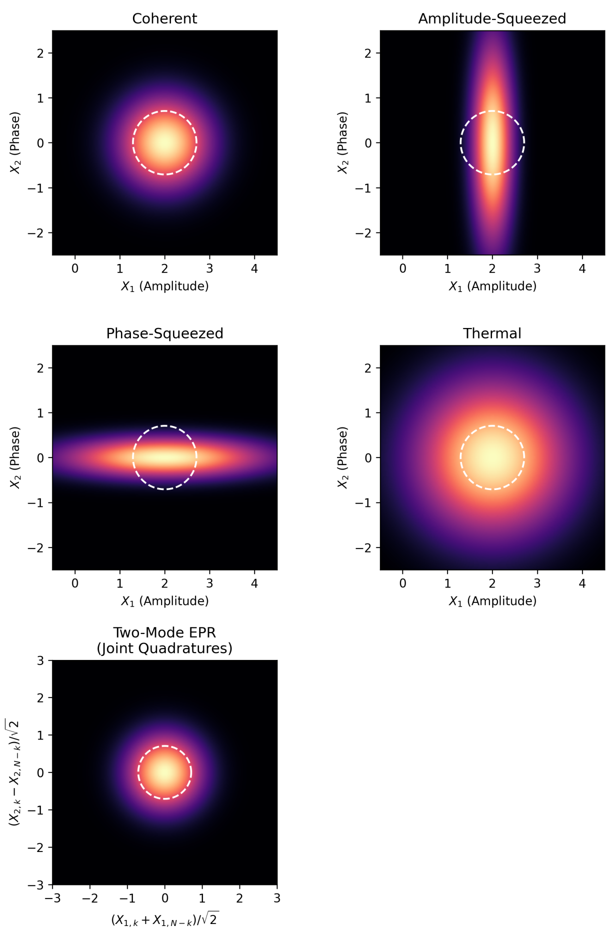

FIG. 3. Wigner phase-space distributions for the five Gaussian input states. The dashed white circle is the vacuum shot-noise reference (Var $= 1/2$). The single-mode geometries (coherent, amplitude-squeezed, phase-squeezed, and thermal) are plotted in the local (X_1, X_2) quadrature basis at coherent displacement $\alpha_k = 2.0$. The two-mode EPR state is plotted in the joint sum/difference quadrature basis, where the squeezing manifests in the correlated macroscopic observables.

function is a circular Gaussian centered at $(\alpha, 0)$, representing the minimum-uncertainty state with no preferred quadrature. This geometry serves as the baseline throughout.

Amplitude-squeezed state. The amplitude quadrature is below the shot-noise level, $\mathrm{Var}(\xi^{(1)}) = e^{-2r}/2$, while the phase quadrature is correspondingly anti-squeezed, $\mathrm{Var}(\xi^{(2)}) = e^{+2r}/2$. The Wigner function is an ellipse elongated along the X_2 axis. Amplitude squeezing narrows the photon-number distribution and suppresses intensity fluctuations at the cost of increased phase uncertainty.

Phase-squeezed state. The roles of the two quadratures are exchanged: $\mathrm{Var}(\xi^{(1)}) = e^{+2r}/2$ and $\mathrm{Var}(\xi^{(2)}) = e^{-2r}/2$. The Wigner ellipse is elongated along the X_1 axis. Because the intensity $|E_k|^2 \approx (X_{1,k} + \alpha_k)^2 + X_{2,k}^2$ receives its dominant noise contribution from the amplitude quadrature when $\alpha_k \neq 0$, phase squeezing broadens the detected intensity distribution and, as the results below show, directly diversifies the nonlinear features generated by FWM.

Two-mode EPR state. Conjugate comb pairs $(k, N{-}1{-}k)$ are prepared in the Einstein–Podolsky–Rosen (EPR) entangled state [13, 17]. The sum quadrature $\xi_k^{(1)} + \xi_{N-1-k}^{(1)}$ and the difference quadrature $\xi_k^{(2)} - \xi_{N-1-k}^{(2)}$ are each squeezed to variance $e^{-2r}/2$, while their conjugate combinations are correspondingly anti-squeezed. Each individual mode, traced over its partner, is a thermal state; the squeezing resides entirely in the inter-mode correlations. This geometry is the paradigmatic two-mode entangled state used in quantum teleportation and dense coding [13, 18].

Thermal state. Both quadratures are broadened by thermal noise with mean occupancy $\bar{n} = \sinh^2 r$, giving $\mathrm{Var}(\xi^{(1,2)}) = \bar{n} + 1/2$. Tying the total photon variance of the thermal state to that of a squeezed state at the same r provides a fair classical-noise control against which quantum squeezing benefits can be assessed. The Wigner function is a circular Gaussian of growing radius, qualitatively distinct from the elongated ellipses of the squeezed geometries.

Kerr four-wave-mixing nonlinear layer

As outlined qualitatively in the introduction, the Kerr nonlinearity induces self- and cross-phase modulation as well as four-wave mixing between comb lines. In this subsection we write the corresponding evolution in a perturbative $\gamma L \ll 1$ model and specify the regime used in our simulations. In a Kerr medium of length L with nonlinear coefficient γ, the instantaneous refractive index of mode k shifts by an amount proportional to the total intensity of all comb lines, producing intensity-dependent phase evolution. For a multi-frequency input with slowly varying envelope amplitudes $\{E_k\}$, the combined effect of self-phase modulation (SPM), cross-phase modulation (XPM), and FWM is treated perturbatively in the limit $\gamma L \ll 1$ [5]. Fig. 4 visualizes the resulting intensity-dependent phase accumulation (panel c) and the cross-intensity product matrix that forms the ELM feature set (panel d). Formally, the output of the Kerr layer is

$$E_k^{\text{out}} = E_k\, e^{i\left(\phi_k^{\text{SPM}} + \phi_k^{\text{XPM}}\right)} + \delta E_k^{\text{FWM}}, \tag{3}$$

where the phase-modulation terms are

$$\phi_k^{\text{SPM}} = \gamma L\, |E_k|^2, \tag{4}$$

$$\phi_k^{\text{XPM}} = 2\gamma L \sum_{l \neq k} |E_l|^2, \tag{5}$$

and the perturbative FWM amplitude correction over energy-conserving triplets is [5]

$$\delta E_k^{\text{FWM}} = -\gamma L\, \mathrm{Im} \sum_{\substack{l-m+n=k \\ l\neq k,\, n\neq k}} E_l E_m^* E_n E_k^*. \tag{6}$$

The condition $l - m + n = k$ is the discrete analogue of $\omega_l - \omega_m + \omega_n = \omega_k$ for a uniformly spaced comb. Only triplets within a bandwidth window of ± 3 modes around k are retained, reflecting the finite phase-matching bandwidth of a realistic fiber or microresonator platform [12]. The Kerr parameter is fixed at $\gamma L = 0.5$, placing the simulation in the moderately nonlinear regime where both SPM/XPM phase modulation and FWM amplitude transfer are significant.

TABLE I. Quadrature variances per mode for each input geometry at squeezing parameter r. The thermal mean occupancy is $\bar{n} = \sinh^2 r$.

State	$\mathrm{Var}(\xi^{(1)})$	$\mathrm{Var}(\xi^{(2)})$
Coherent	$\frac{1}{2}$	$\frac{1}{2}$
Amplitude-squeezed	$\frac{e^{-2r}}{2}$	$\frac{e^{+2r}}{2}$
Phase-squeezed	$\frac{e^{+2r}}{2}$	$\frac{e^{-2r}}{2}$
Two-mode EPR	correlated pairs $(k,\ N{-}1{-}k)$; see text	
Thermal	$\bar{n} + \frac{1}{2}$	$\bar{n} + \frac{1}{2}$

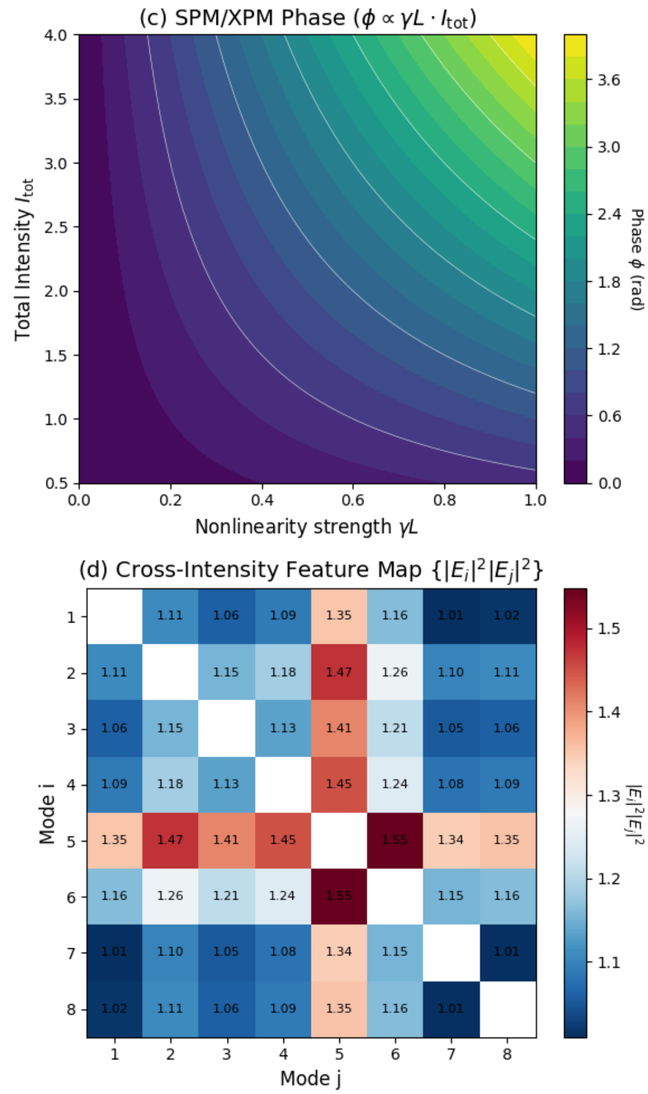

FIG. 4. (c) Intensity-dependent phase $\phi = \gamma L \cdot I_{\text{total}}$ accumulated by a comb line as a function of nonlinearity strength γL and total comb intensity I_{total}; contours mark equal phase accumulation. (d) Cross-intensity product matrix $\{|E_i|^2|E_j|^2\}$ for $N = 8$ representative comb modes (diagonal entries excluded). The spread of off-diagonal values quantifies the number of independent polynomial features available to the linear readout and directly determines the effective rank d_{eff} of the Gram matrix (Eq. (9)).

Feature extraction and linear readout

The hidden-layer feature vector is defined as

$$\mathbf{h} = \Big[|E_k^{\text{out}}|^2,\ \mathrm{Re}(E_k^{\text{out}}),\ \mathrm{Im}(E_k^{\text{out}}),\ |E_i^{\text{out}}|^2 |E_j^{\text{out}}|^2\Big]_{k;\,(i,j)}, \tag{7}$$

concatenating N intensity readouts, $2N$ quadrature readouts, and up to 50 pairwise intensity cross-products. Intensity readouts correspond to direct photodetection; quadrature readouts are obtained by balanced homodyne detection; pairwise products correspond to FWM-generated sidebands measurable as optical beat tones. This feature set captures the first- and second-order moments of the Kerr-transformed field, forming a degree-four polynomial kernel in the input amplitudes.

Input samples $\mathbf{x} \in \mathbb{R}^d$ are projected onto N modes by a fixed random matrix $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d\times N}$ with unit-norm columns, giving the per-mode displacement $\alpha_k = \mu_0[\mathbf{x}\mathbf{W}_{\text{in}}]_k$ with mean field amplitude $\mu_0 = 2.0$. Quantum noise $\boldsymbol{\xi}$ drawn from the appropriate Gaussian state is added to each mode before the Kerr layer, as in Eq. (2). The matrix $\mathbf{W}_{\text{in}}$ is drawn once and held fixed across all values of r and all states, so that performance differences are attributable solely to the noise geometry.

A ridge-regression readout [19]

$$\hat{y} = \mathbf{W}_{\text{out}}\mathbf{h} + b, \qquad \mathbf{W}_{\text{out}} = (\mathbf{H}^\top\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}^\top\mathbf{Y}, \quad (8)$$

is trained on a 70% split with regularization $\lambda = 10^{-2}$ and evaluated on the remaining 30%. For classification tasks $\hat{y}$ is the one-hot class probability vector and the score is test accuracy; for regression (NARMA-10) the score is $1 - \text{NRMSE}$, so that higher values are always better.

Learning-capacity diagnostics

Two kernel-based diagnostics characterize why one noise geometry outperforms another, decoupling the question of how much information the features contain from how well that information is aligned to the task structure.

The *effective rank* of the Gram matrix $\mathbf{K} = \mathbf{H}\mathbf{H}^\top$ is defined via the Shannon entropy of its normalized eigenspectrum [20]:

$$d_{\text{eff}} = \exp\left(-\sum_i \tilde{\lambda}_i \ln \tilde{\lambda}_i\right), \qquad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (9)$$

The quantity d_{eff} counts the number of effectively independent feature dimensions: it equals N when the eigenspectrum is flat and approaches 1 when a single eigenvalue dominates. For a reservoir computer to generalize, the feature kernel must span enough independent dimensions to separate the classes or time-series values of interest, so d_{eff} serves as a task-independent proxy for representational power.

The *centered kernel alignment* (CKA) is a normalized inner product between the feature kernel and the ideal task kernel [21]:

$$\text{CKA}(\mathbf{K}, \mathbf{Y}) = \frac{\langle \mathbf{K}_c, \mathbf{Y}_c\rangle_F}{\|\mathbf{K}_c\|_F\,\|\mathbf{Y}_c\|_F}, \quad (10)$$

where $\mathbf{K}_c$ and $\mathbf{Y}_c$ are the doubly-centered kernel matrices [21]. For classification, $\mathbf{Y}$ is the one-hot label matrix; for regression, $\mathbf{Y}$ is the target vector. CKA measures how well the geometry of the feature space matches that required by the task. Two feature spaces of equal d_{eff} can differ substantially in CKA if their dominant dimensions are orthogonal to the task's separating hyperplane. Together, d_{eff} and CKA provide a mechanistic decomposition of task score: improvements from increased dimensionality appear in d_{eff}, while improvements from better feature-task alignment appear in CKA.

RESULTS

Benchmark tasks

The ELM is evaluated on three tasks of increasing structural difficulty. (i) *Iris (linear)*: multiclass classification of the UCI Iris dataset ($n = 150$, 4 features, 3 classes) [22], which is largely linearly separable and serves as a check that no geometry catastrophically impairs performance. (ii) *XOR Spiral (nonlinear)*: binary classification of a 2-D spiral dataset ($n = 500$) in which the two classes are interleaved and the decision boundary is radially nonlinear, requiring genuine nonlinear feature extraction. (iii) *NARMA-10 (memory)*: ten-step nonlinear autoregressive moving-average regression [23] ($n = 790$), which requires the reservoir to retain information over ten time steps simultaneously and is the canonical benchmark for memory capacity in reservoir computing [1].

All results are averaged over $N_{\text{seed}} = 5$ Monte Carlo seeds. All results at $r = 0$ coincide for all states (the coherent limit) and serve as a single baseline.

Task-performance sweep

Fig. 5 presents the full 3×3 grid of task score, effective rank d_{eff}, and CKA as functions of r for $N = 32$ modes.

On the Iris task, all geometries perform comparably near the coherent ceiling of 0.26 accuracy. The near-linear separability of this dataset reduces the benefit of enhanced feature nonlinearity; the small improvement observed for two-mode EPR squeezing at $r = 0$ is a finite-sample artifact.

On the XOR Spiral task, both phase-squeezed and thermal states reach $\sim$0.52 at large r, a 16% improvement over the coherent baseline of 0.45. The improvement is non-monotone for some geometries, reflecting competition between the increased noise floor at large r and the richer Wigner-function structure exploited by FWM.

The most pronounced result appears on NARMA-10. At $r = 2$, phase-squeezed inputs reach $1 - \text{NRMSE} = -0.31$, compared to -0.58 for the coherent baseline, a 27.5% absolute improvement. Amplitude-squeezed and thermal states show intermediate gains, while two-mode

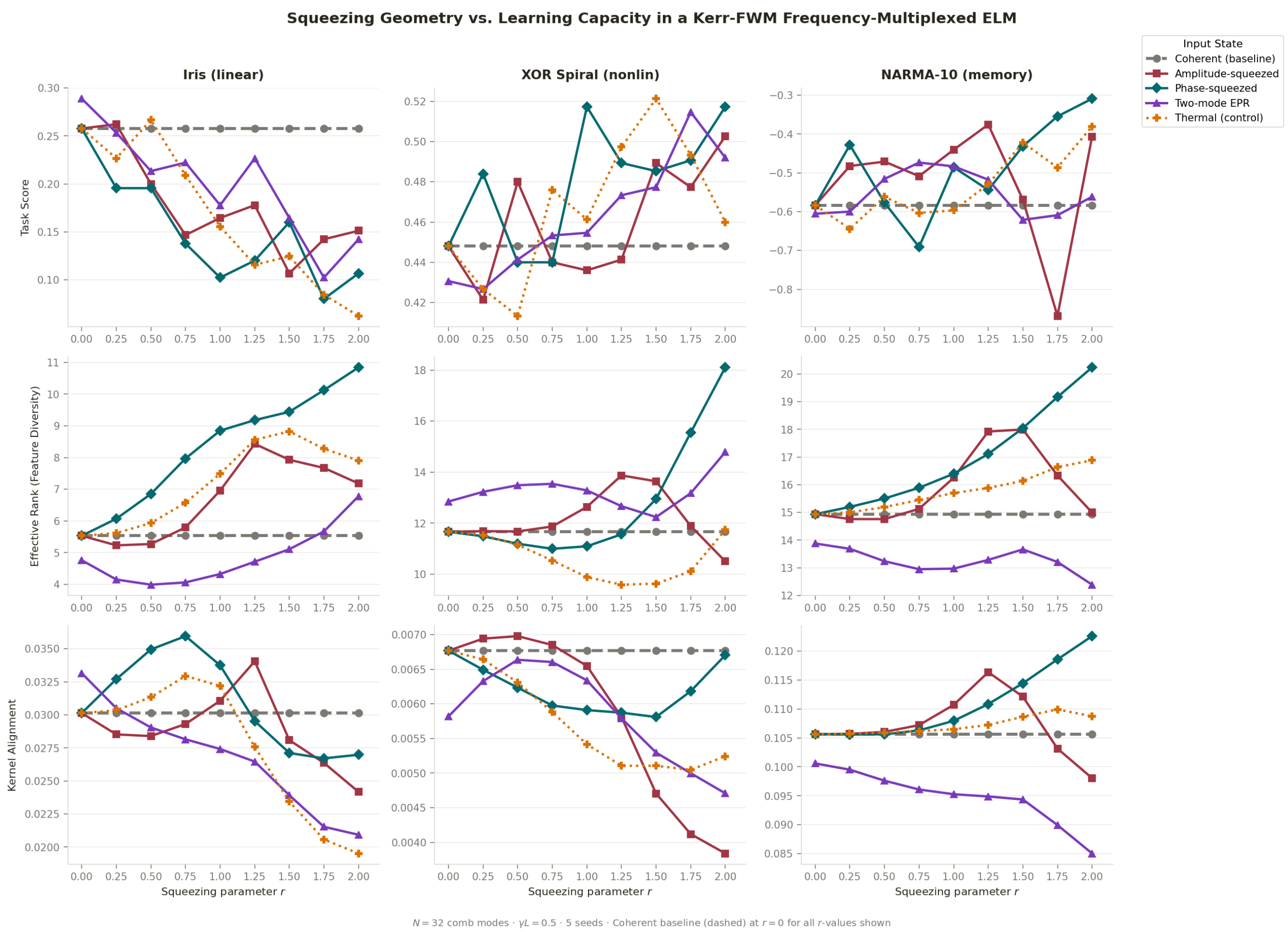

FIG. 5. Full benchmark sweep for $N = 32$ modes and $\gamma L = 0.5$. Each panel shows one metric (task score, effective rank d_{eff}, or CKA) for one task as a function of squeezing parameter r. The coherent baseline (gray dashed) is independent of r by construction. Results are averaged over 5 random seeds.

EPR squeezing provides the smallest improvement despite carrying the largest total fluctuations.

Feature diversity and kernel alignment

The middle and bottom rows of Fig. 5 expose the mechanism behind the task-score trends. For NARMA-10, phase-squeezed inputs raise d_{eff} from the coherent baseline of $\approx$15 to $\approx$20 at $r = 2$, a 33% increase. This rank increase follows from the expansion of the amplitude-quadrature variance $\sigma^2_{X_1} = e^{+2r}/2$, which broadens the distribution of $|E_k|^2$ and generates higher-order intensity moments that populate additional singular-value dimensions in $\mathbf{K}$.

Two-mode EPR squeezing produces the opposite effect. The inter-mode correlations it imposes cause conjugate pairs $(k, N{-}1{-}k)$ to fluctuate in concert, reducing the number of statistically independent directions in $\mathbf{K}$ to $\approx$12 at large r, a 20% reduction below the coherent baseline. The CKA follows the same trend: phase-squeezed inputs reach CKA ≈ 0.12 on NARMA-10 at $r = 2$, while two-mode EPR inputs fall to 0.085, indicating that EPR correlations misalign the feature kernel from the temporal structure of the task.

Optimal-geometry phase diagram

Fig. 6 shows which input geometry achieves the highest task score at each value of r for each benchmark. On Iris, the coherent baseline remains competitive throughout. On XOR Spiral, phase-squeezed and thermal states alternate. On NARMA-10, phase-squeezed inputs lead at every $r \geq 0.25$ without exception, establishing a robust advantage for memory-intensive tasks.

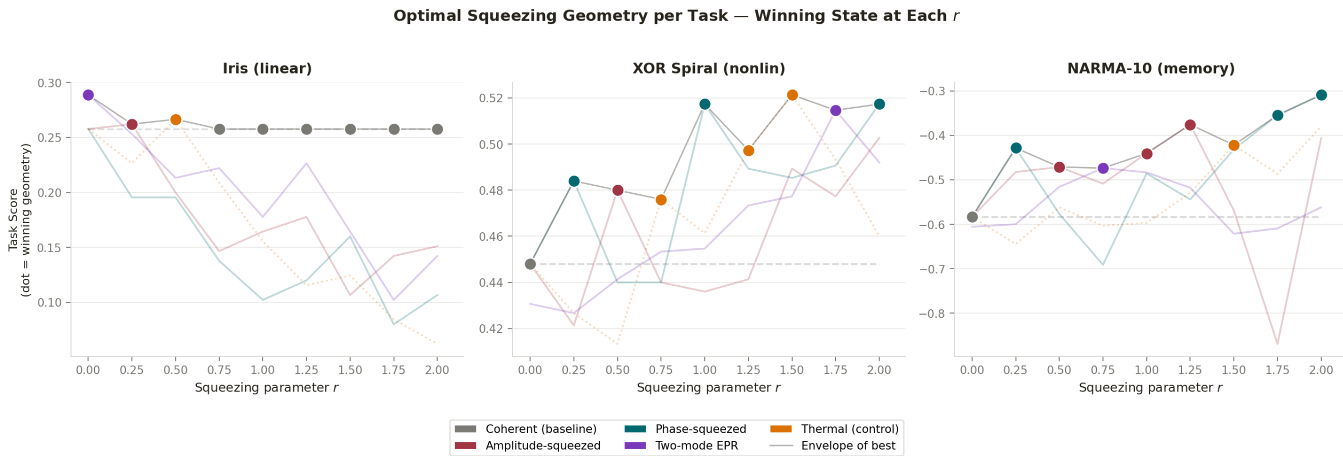

FIG. 6. Optimal squeezing geometry per task. Each colored dot indicates which input state achieves the highest task score at that value of r. The envelope (thin gray line) traces the best achievable performance across all geometries. Phase-squeezed inputs (teal) dominate the NARMA-10 panel for all $r \geq 0.25$.

DISCUSSION

The central result of this study is a mechanistically transparent link between the orientation of the quantum noise ellipse and the learning capacity of a Kerr-FWM ELM. Phase squeezing expands the amplitude-quadrature variance, which broadens the photon-number distribution and diversifies the polynomial features produced by FWM. This phase-space deformation is qualitatively distinct from the gain achieved with a thermal state of identical total photon variance: thermal noise broadens both quadratures equally and degrades the signal-to-noise ratio of the coherent displacement, whereas phase squeezing concentrates the additional noise in the quadrature that directly drives photon-number fluctuations and, through them, the FWM intensity cross-products.

Two-mode EPR squeezing represents an instructive counterexample. Although it is the paradigmatic resource for quantum communication and quantum teleportation [13, 18], it introduces classical inter-mode correlations that reduce feature independence. This illustrates a general principle: quantum-optical entanglement does not generically benefit machine learning; the correlation structure of the entanglement must match the architecture of the nonlinear mixer. Whether cluster-state multimode entanglement, in which modes are entangled in a graph that mirrors the FWM coupling topology, could overcome this limitation is an interesting direction for future study [16].

The present model operates in the perturbative Kerr regime and uses a classical linear readout, placing it squarely within the reach of current fiber- and microresonator-based platforms [3, 11, 12]. Several extensions remain for future work. First, at strong nonlinearity ($\gamma L \gtrsim 1$), the perturbative FWM expansion breaks down and the dynamics become chaotic; squeezed-state benefits may be amplified or suppressed in this regime. Second, non-Gaussian states (Fock, cat, and GKP states) could in principle produce feature distributions inaccessible to any Gaussian input; the experimental cost-benefit tradeoff of generating such states would need assessment. Third, photon loss in integrated-photonic waveguides progressively degrades pure squeezed inputs toward mixed thermal states; a quantitative analysis of loss-robustness is required before experimental implementation.

CONCLUSION

We have demonstrated numerically that the quantum noise geometry of a Gaussian-state optical frequency comb input substantially affects the learning capacity of a Kerr-FWM ELM. Phase-squeezed inputs increase the effective rank of the feature Gram matrix by up to 33% and the centered kernel alignment with the task structure, yielding a 27.5% improvement in 1−NRMSE over the coherent baseline on the NARMA-10 memory benchmark at $r = 2$. The improvement scales with task memory depth: linearly separable tasks are largely unaffected, while time-series regression benefits strongly. Two-mode EPR squeezing suppresses feature diversity and should be avoided in this architecture. These results establish that quantum state engineering of photonic reservoir inputs, now a routine capability in integrated quantum photonics laboratories, is a viable route to enhanced machine learning performance without modification to the nonlinear hardware.

The author thanks Dr. Harry Levine from the Department of Physics at the University of California, Berkeley, for his guidance and teachings that facilitated and motivated this study. The simulation code was developed with NumPy [24], scikit-learn [25], and Matplotlib [26]. The jupyter notebook can be found here.

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